3.13 Chain Conditions

Definition 3.13.1 (Noetherian / Artinian / Finite Modules)
*We say an A-module M is **Noetherian** if it satisfies the **ascending chain condition**, namely any ascending chain of submodules*

$$M_0 \subseteq M_1 \subseteq \cdots \subseteq M$$

eventually stabilizes, i.e $M_n = M_{n+1}$ for all $n \geq N$.

*Similarly we say an A-module M is **Artinian** if it satisfies the **descending chain condition**, namely any descending chain of submodules*

$$M \supseteq M_0 \supseteq M_1 \supseteq \cdots$$

eventually stabilizes.

Definition 3.13.2 (Noetherian / Artinian Ring)
*We say a ring A is **Noetherian** (resp. Artinian) if it is Noetherian (resp. Artinian) as an A-module.*

The following is useful

Proposition 3.13.3 (Noetherian criterion)
Let M be an A-module. The following are equivalent

a) M is Noetherian
b) Every submodule $N \subseteq M$ is finitely-generated
c) Every set of submodules has a maximal element

Proposition 3.13.4
Let M be an A-module and A a Noetherian ring. Then M is Noetherian if and only if it is finitely generated as an A-module.

Proposition 3.13.5 (Restriction of Scalars preserves finiteness)
Let $\phi : A \to B$ be a finite A-algebra and M a (3.3.89) B-module. Then $[M]_\phi$ is a (3.3.89) A-module.

Proof. We suppose that M is generated by $m_1, ..., m_n$, and B is generated by $b_1, ..., b_m$. Then we claim that the elements $b_i m_j$ generate $[M]_\phi$. ∎

Proposition 3.13.6
Let A be a Noetherian ring and $\mathfrak{a} \triangleleft A$ an ideal. Then $A/\mathfrak{a}$ is Noetherian.

Proof. Consider an increasing sequence of ideals $\mathfrak{a}_i \triangleleft A/\mathfrak{a}$. Then by (3.3.56) this corresponds to an increasing sequence of ideals $\mathfrak{a}_i' \triangleleft A$ containing $\mathfrak{a}$. As A is Noetherian, this sequence eventually stabilizes. Again by (3.3.56) the original sequence stabilizes. ∎

Proposition 3.13.7 (Localisation preserves Noetherian)
Let A be a Noetherian ring and $S \subset A$ a multiplicative set. Then $S^{-1}A$ is Noetherian.

Proof. Let $\mathfrak{b}_0 \subseteq \mathfrak{b}_1 \subseteq \cdots$ be an ascending chain of ideals of $S^{-1}A$. Contracting gives an ascending chain of ideals $\mathfrak{b}_0^c \subseteq \mathfrak{b}_1^c \subseteq \cdots$ of A, which stabilizes as A is Noetherian. As every ideal of $S^{-1}A$ is extended (3.5.18), $\mathfrak{b}_i = \mathfrak{b}_i^{ce}$ for each i, so the original chain stabilizes too. ∎

Proposition 3.13.8
Let A be a Noetherian ring then every finitely-generated A-algebra is Noetherian.

In particular $A[X_1, ..., X_n]$ is Noetherian.

Proof. By (3.13.6) it's enough to show that $A[X_1, ..., X_n]$ is Noetherian. By induction and (3.10.4) it's enough to consider the case $n = 1$. Let $\mathfrak{a} \triangleleft A[X]$ then by (3.13.3) it's enough to show that $\mathfrak{a}$ is finitely-generated.

Define

$$\tilde{\mathfrak{a}}_i := \{\ell(f) \mid f \in \mathfrak{a} \text{ s.t. } \deg(f) = i\}$$

Then clearly $\tilde{\mathfrak{a}}_i \triangleleft A$ is an ideal. This is an increasing sequence of ideals, and so it stabilizes for $i \geq d$ for some $d > 0$. Furthermore each ideal is finitely generated

$$\tilde{\mathfrak{a}}_i := \left(c_{i1}, ..., c_{in(i)}\right)$$

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where $c_{ij} = \ell(f_{ij})$ for polynomials $f_{ij} \in \mathfrak{a}$ of degree i. We claim that

$$\mathfrak{a} = (f_{ij})_{i \leq d; j \leq n(i)}$$

Denote the right hand side by $\mathfrak{b}$, then clearly $\mathfrak{b} \subseteq \mathfrak{a}$. We show by induction on $m = \deg(f)$ that $f \in \mathfrak{a} \implies f \in \mathfrak{b}$. Let $m' \coloneqq \min(m, d)$. Then by assumption $\ell(f) \in \tilde{\mathfrak{a}}_m = \tilde{\mathfrak{a}}_{m'}$ so there exists $\lambda_j \in A$ such that

$$\ell(f) = \sum_{j=1}^{n(m')} c_{m'j} \lambda_j$$

Consider the decomposition

$$f = \left(f - \sum_{j=1}^{n(m')} \lambda_j f_{m'j} X^{m-m'} \right) + \sum_{j=1}^{n(m')} \lambda_j f_{m'j} X^{m-m'}$$

The first term has strictly smaller degree than f, so by the inductive hypothesis lies in $\mathfrak{b}$. Therefore $f \in \mathfrak{b}$ as required.■

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